MuTrans Analysis of Epithelial-Mesenchymal Transition (EMT)

Load raw dataset and explore with Scanpy

In [1]:
```
import pandas as pd
import scanpy as sc
sc.settings.set_figure_params(dpi=100, frameon=False, figsize=(3, 3), facecolor='whit
e')

import matplotlib.pyplot as plt
import seaborn as sns
color_palette = sns.color_palette('Set1', 5)

import pyMuTrans as pm

datadir = "../Data/"
adata = sc.read_csv(datadir+'GSE110357_htseq_counts_all_v1.csv.gz').T
```

```
/Users/cliffzhou/opt/anaconda3/lib/python3.7/site-packages/pyemma/__init__.py:92: Us
erWarning: You are not using the latest release of PyEMMA. Latest is 2.5.7, you have
2.5.6.
  .format(latest=latest, current=current), category=UserWarning)
```

In [2]:
```
import mygene
adata.var_names =[x.split(".")[0] for x in adata.var_names]
mg = mygene.MyGeneInfo()
geneList = adata.var_names
geneSyms = mg.querymany(geneList , scopes='ensembl.gene', fields='symbol', species='m
ouse')
```

```
querying 1-1000...done.
querying 1001-2000...done.
querying 2001-3000...done.
querying 3001-4000...done.
querying 4001-5000...done.
querying 5001-6000...done.
querying 6001-7000...done.
querying 7001-8000...done.
querying 8001-9000...done.
querying 9001-10000...done.
querying 10001-11000...done.
querying 11001-12000...done.
querying 12001-13000...done.
querying 13001-14000...done.
querying 14001-15000...done.
querying 15001-16000...done.
querying 16001-17000...done.
querying 17001-18000...done.
querying 18001-19000...done.
querying 19001-20000...done.
querying 20001-21000...done.
querying 21001-22000...done.
querying 22001-23000...done.
querying 23001-24000...done.
querying 24001-25000...done.
querying 25001-26000...done.
querying 26001-27000...done.
querying 27001-28000...done.
querying 28001-29000...done.
querying 29001-30000...done.
querying 30001-31000...done.
querying 31001-32000...done.
querying 32001-33000...done.
querying 33001-34000...done.
querying 34001-35000...done.
querying 35001-36000...done.
querying 36001-37000...done.
querying 37001-38000...done.
querying 38001-39000...done.
querying 39001-40000...done.
querying 40001-41000...done.
querying 41001-42000...done.
querying 42001-43000...done.
querying 43001-44000...done.
querying 44001-45000...done.
querying 45001-46000...done.
querying 46001-47000...done.
querying 47001-48000...done.
querying 48001-49000...done.
querying 49001-49677...done.
Finished.
1 input query terms found dup hits:
        [('ENSMUSG00000100417', 2)]
3700 input query terms found no hit:
        ['ENSMUSG00000000325', 'ENSMUSG00000004613', 'ENSMUSG00000011052', 'ENSMUSG0
0000021745', 'ENSMUSG000
Pass "returnall=True" to return complete lists of duplicate or missing query terms.
```

In [3]:
```
gene_name = [x['symbol'] if x.__contains__('symbol') else x['query'] for x in geneSym
s if not x['query']=='ENSMUSG00000100417']
adata = adata [:,adata.var_names !='ENSMUSG00000100417']
adata.var_names = gene_name
```

In [4]:
```
adata.var_names_make_unique()
sc.pp.filter_cells(adata, min_genes=200)
sc.pp.filter_genes(adata, min_cells=3)
```

In [5]:
```
adata.var['ERCC'] = adata.var_names.str.startswith('ERCC')  # annotate the group of m
itochondrial genes as 'mt'
sc.pp.calculate_qc_metrics(adata, qc_vars=['ERCC'], percent_top=None, log1p=False, in
place=True)
```

In [6]:
```
adata.obs['n_counts']=adata.X.sum(axis=1)
adata = adata[adata.obs['n_counts']>100000,:]
```

In [7]:
```
adata = adata[adata.obs['pct_counts_ERCC']<2,:]
```

In [8]:
```
adata.raw = adata
genes = pd.read_csv(datadir+'emt_genes.csv')
genes_intersect = list(set(genes['genes'].to_list()) & set(gene_name))
adata = adata[:,adata.var_names.isin(genes_intersect)]
```

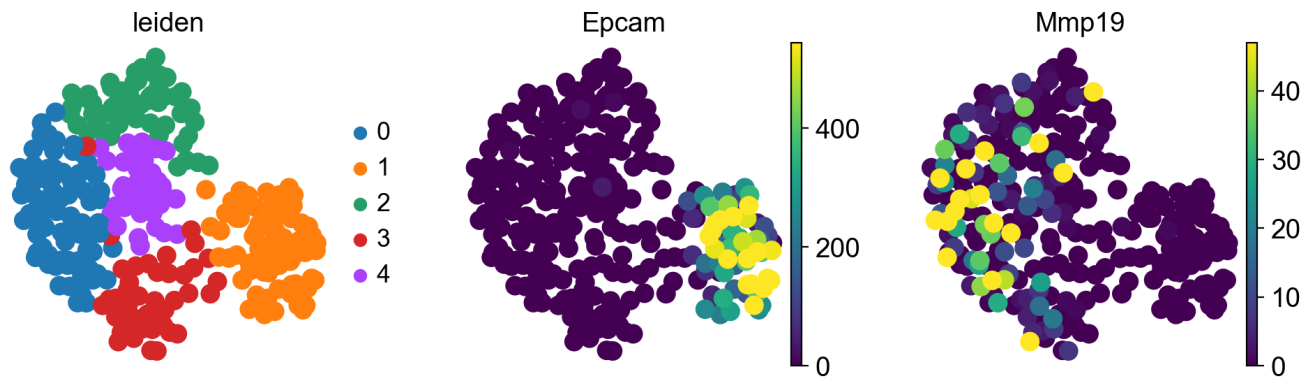
In [9]:
```
sc.pp.log1p(adata)
sc.pp.scale(adata)
```

```
/Users/cliffzhou/opt/anaconda3/lib/python3.7/site-packages/scanpy/preprocessing/_sim
ple.py:339: UserWarning: Revieved a view of an AnnData. Making a copy.
  view_to_actual(adata)
```

In [10]:
```
sc.tl.pca(adata, svd_solver='arpack')
sc.pp.neighbors(adata,metric = 'cosine',n_neighbors=50, use_rep='X')
sc.tl.umap(adata)
```

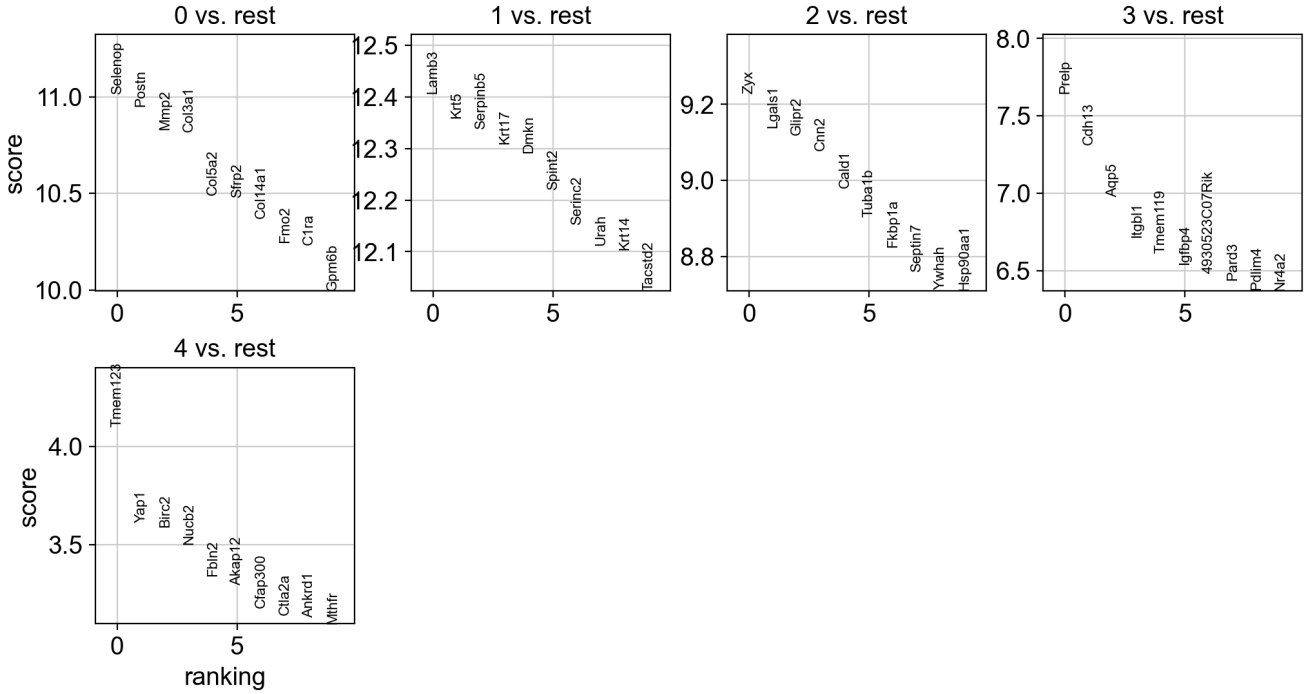
Using leiden algorithm to detect clusters

In [11]:
```
sc.settings.set_figure_params(dpi=100, frameon=False, figsize=(3, 3), facecolor='whit
e')
sc.tl.leiden(adata,resolution = 1.0)
sc.pl.umap(adata, color=['leiden','Epcam','Mmp19'], vmax = 'p95')
```

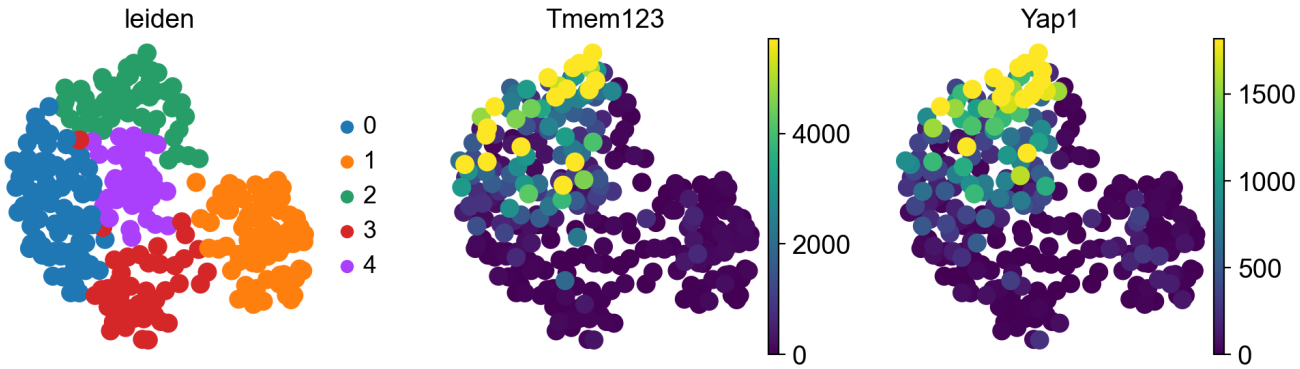


In [12]:
```
sc.tl.rank_genes_groups(adata, 'leiden', method='wilcoxon')
sc.pl.rank_genes_groups(adata, n_genes=10, sharey=False)
```

```
/Users/cliffzhou/opt/anaconda3/lib/python3.7/site-packages/scanpy/tools/_rank_genes_
groups.py:404: RuntimeWarning: overflow encountered in expm1
  foldchanges = (self.expm1_func(mean_group) + 1e-9) / (
/Users/cliffzhou/opt/anaconda3/lib/python3.7/site-packages/scanpy/tools/_rank_genes_
groups.py:405: RuntimeWarning: overflow encountered in expm1
  self.expm1_func(mean_rest) + 1e-9
/Users/cliffzhou/opt/anaconda3/lib/python3.7/site-packages/scanpy/tools/_rank_genes_
groups.py:405: RuntimeWarning: invalid value encountered in true_divide
  self.expm1_func(mean_rest) + 1e-9
/Users/cliffzhou/opt/anaconda3/lib/python3.7/site-packages/scanpy/tools/_rank_genes_
groups.py:408: RuntimeWarning: divide by zero encountered in log2
  foldchanges[global_indices]
```

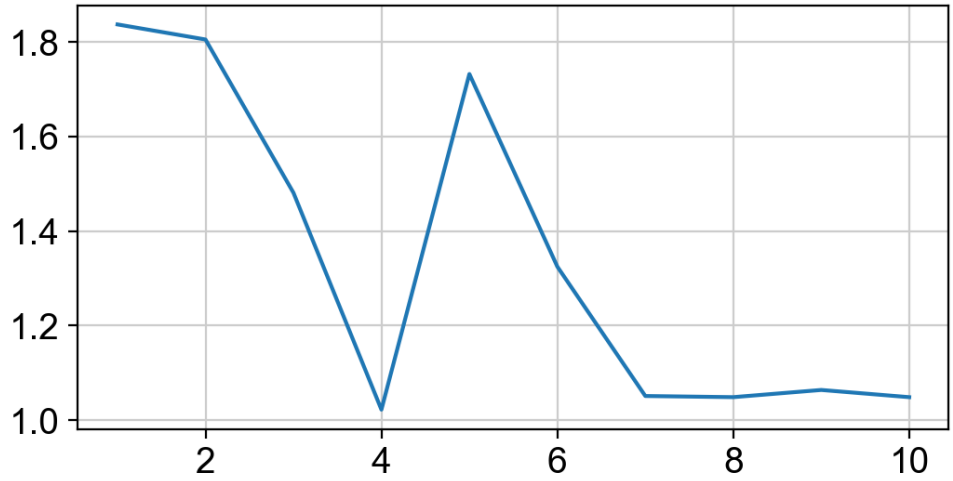


In [13]:
```
sc.pl.umap(adata, color=['leiden','Tmem123','Yap1'], vmax = 'p95')
```



Estimate Cluster Number with EPI (Eigen-Peak Index)

In [14]:
```
fig = plt.figure(figsize=(6, 3))
par = {"choice_distance":"cosine"}
out = pm.plot_cluster_num(adata, par, k_plot= 10) #look for the peaks
```

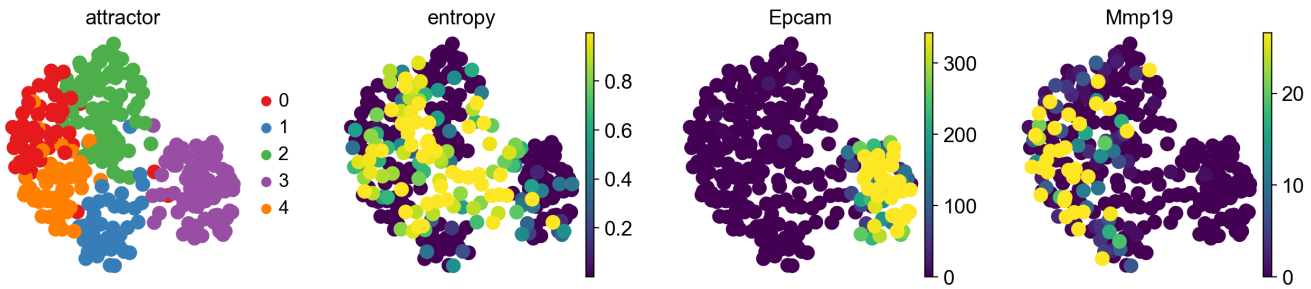


MuTrans Analysis Based on the AnnData Object

In [15]:
```
par = {"choice_distance":"cosine","perplex":150.0,"K_cluster":5.0,"trials":10,"reduct
ion_coord":'umap',"write_anndata":True}
adata = pm.dynamical_analysis(adata,par)
```

In [16]:
```
sc.pl.umap(adata, color=['attractor','entropy','Epcam','Mmp19'], vmax = 'p90', palett
e = color_palette)
```

```
... storing 'attractor' as categorical
```



In [17]:
```
sc.tl.rank_genes_groups(adata, 'attractor', method='wilcoxon')
sc.pl.rank_genes_groups(adata, n_genes=10, sharey=True, ncols = 5)
```

```
/Users/cliffzhou/opt/anaconda3/lib/python3.7/site-packages/scanpy/tools/_rank_genes_
groups.py:404: RuntimeWarning: overflow encountered in expm1
  foldchanges = (self.expm1_func(mean_group) + 1e-9) / (
/Users/cliffzhou/opt/anaconda3/lib/python3.7/site-packages/scanpy/tools/_rank_genes_
groups.py:405: RuntimeWarning: overflow encountered in expm1
  self.expm1_func(mean_rest) + 1e-9
/Users/cliffzhou/opt/anaconda3/lib/python3.7/site-packages/scanpy/tools/_rank_genes_
groups.py:405: RuntimeWarning: invalid value encountered in true_divide
  self.expm1_func(mean_rest) + 1e-9
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groups.py:408: RuntimeWarning: divide by zero encountered in log2
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```
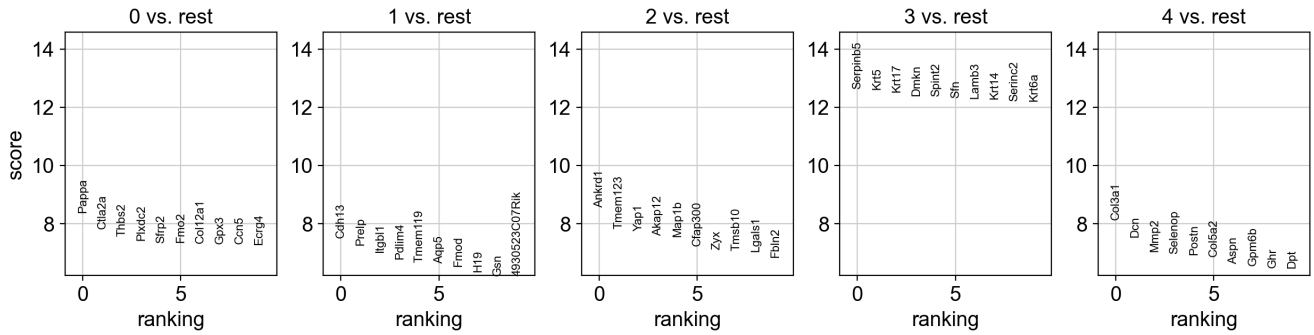
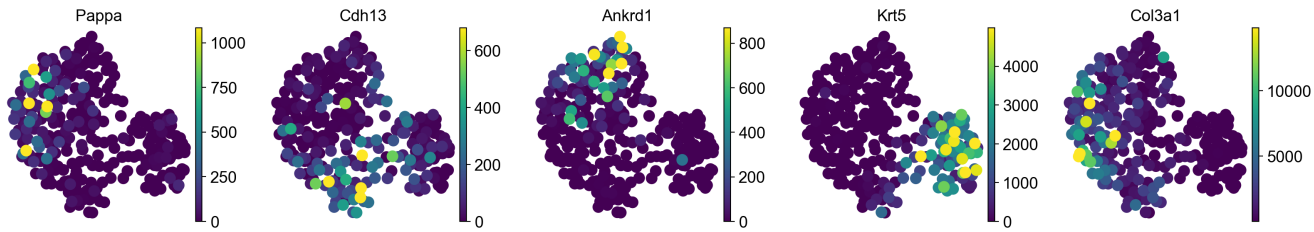



In [18]:
```
sc.pl.umap(adata, color=['Pappa','Cdh13','Ankrd1','Krt5','Col3a1'],vmax = 'p99',vmin
= 'p1', ncols = 5)
```



In [19]:
```
sc.settings.set_figure_params(frameon=False,figsize=(4, 3))
sc.pl.violin(adata, keys = ['Epcam','Krt5','Zeb2','Prrx1','Mmp19','Aspn'], groupby =
'attractor')
```

```
/Users/cliffzhou/opt/anaconda3/lib/python3.7/site-packages/seaborn/_decorators.py:4
3: FutureWarning: Pass the following variable as a keyword arg: x. From version 0.1
2, the only valid positional argument will be `data`, and passing other arguments wi
thout an explicit keyword will result in an error or misinterpretation.
  FutureWarning
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/Users/cliffzhou/opt/anaconda3/lib/python3.7/site-packages/seaborn/_decorators.py:4
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  FutureWarning
```
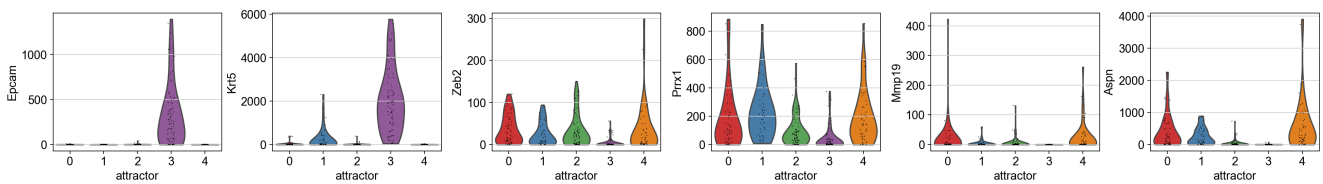
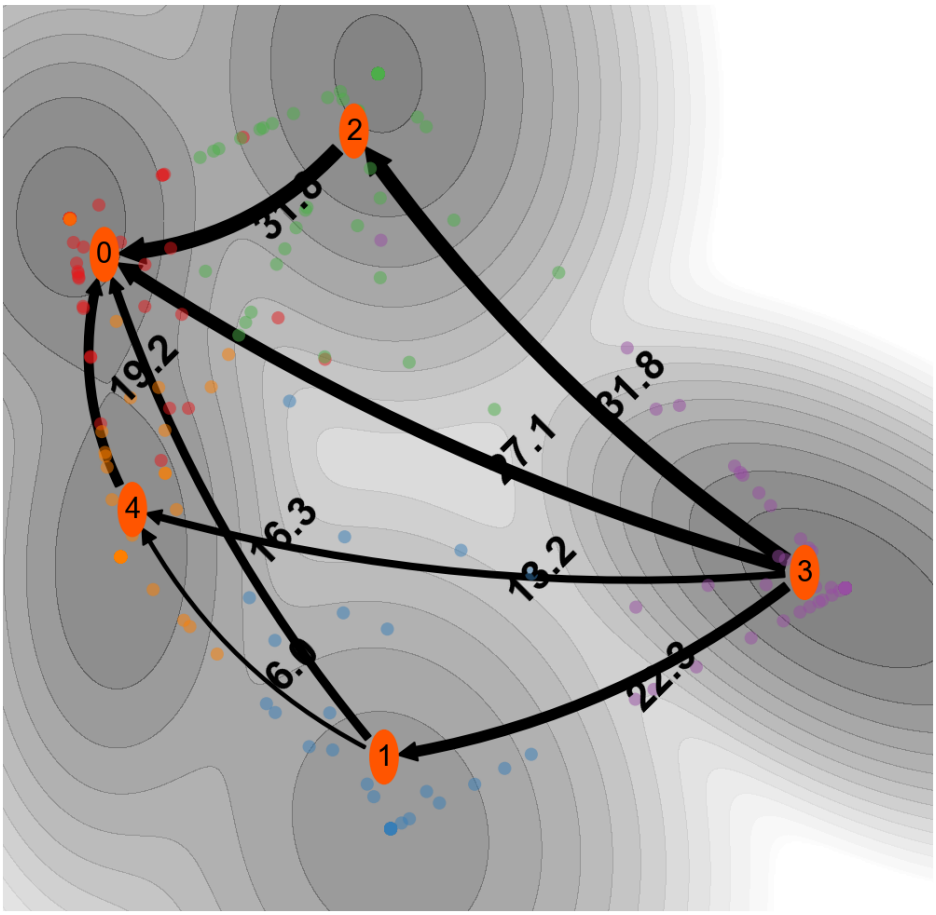
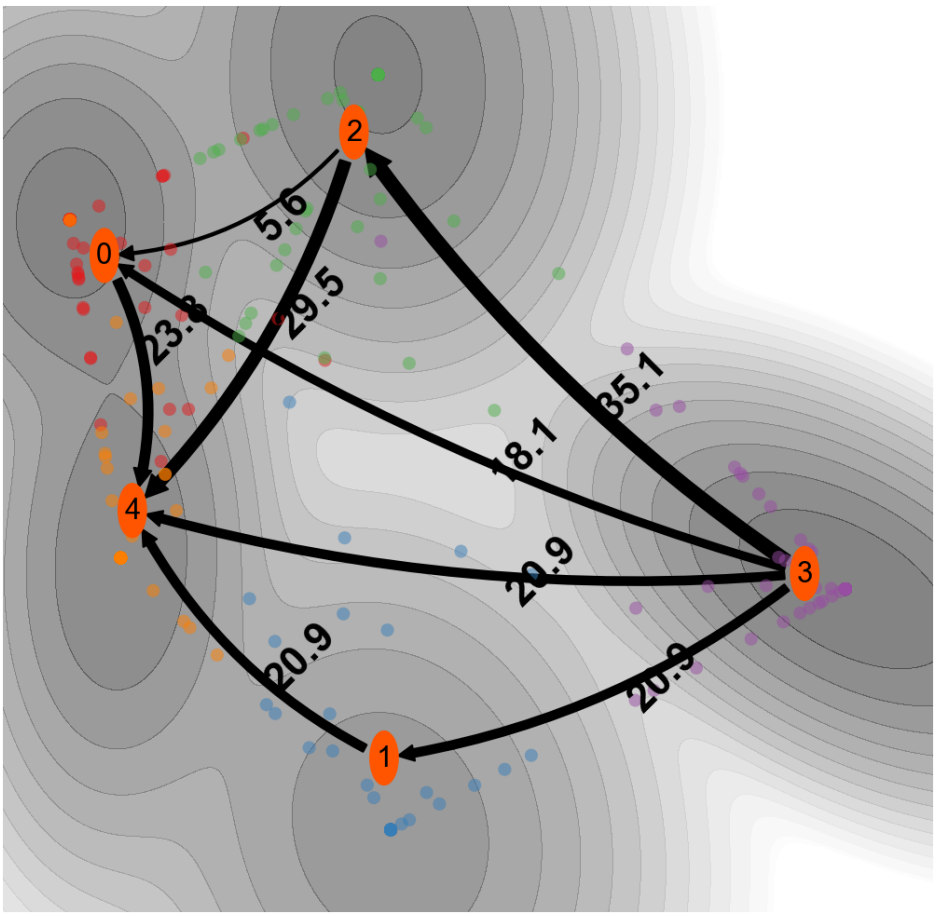



Infer the transition trajectories from Epithelial to Mesenchymal, with the MPPT (most probable path tree) approach

In [20]:
```
fig = plt.figure(figsize=(8, 8))
pm.infer_lineage(adata,si=3,sf=0,method = "MPPT",size_point =40,alpha_point = 0.5)
```



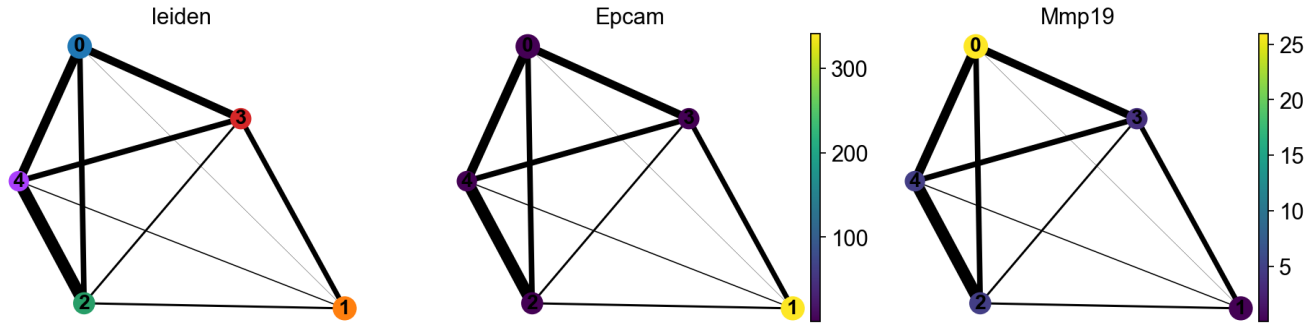
In [21]:
```
fig = plt.figure(figsize=(8, 8))
pm.infer_lineage(adata,si=3,sf=4,method = "MPPT",size_point =40,alpha_point = 0.5)
```



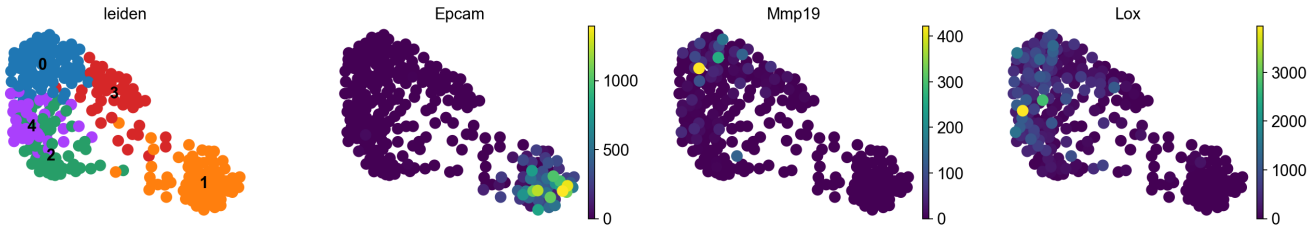
PAGA Analysis

In [22]:
```
sc.tl.paga(adata, groups='leiden')
```

In [23]:
```
sc.pl.paga(adata, color=['leiden','Epcam','Mmp19'])
```



In [24]:
```
sc.tl.draw_graph(adata, init_pos='paga')
sc.pl.draw_graph(adata, color=['leiden','Epcam','Mmp19','Lox'], legend_loc='on data')
```



In [25]:
```
from scipy.io import savemat
mdic = {"data": adata.X, "genes": adata.var_names.to_numpy()}
savemat('emt_data.mat',mdic)
```